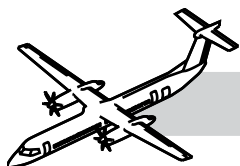


CHAPTER 20

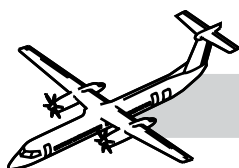
STANDARD PRACTICE

CONTENTS

	Page
20-00-00 STANDARD PRACTICE	20-1
20-00-01 MARKING OF THE AIRCRAFT PARTS AND MATERIALS	20-1
Introduction	20-1
General	20-2
Marking of the Aircraft Parts and Materials with Markers.....	20-3
Cleaning the Drain Hole on the Pitot Static Probes	20-3
Inspection of Seven-by-Nineteen and Seven-by-Seven Control Cables	20-5
Guidelines for Rigid Line Repairs.....	20-7
Permanent Repairs	20-7
Task Removal & Installation of Electrostatic Discharge Sensitive Devices.....	20-11
Metal Case	20-11
Printed Circuit Board	20-11
Cleaning of Electrical Connectors, Switches or Relays.....	20-13
Cleaning of the Wiring Harness Connectors (Engine).....	20-13
Demagnetization of the Aircraft Components and/or Structures	20-15
Replacement of the Lens Cap Assembly and Switch Light Bulb.....	20-15
Electrical Test of the Aircraft Wiring.....	20-17
Installation of the Aircraft Antennas & Application of Sealant to Antenna Connectors.....	20-17
Removal/Installation of the Instrument Panel Integral Lamp Assembly	20-17
Vacuum Anchor System	20-17



	Page
20-30-00 REMOVAL OF ELECTROSTATIC DISCHARGE SENSITIVE DEVICES.....	20-19
Introduction	20-19
General	20-19
20-00-00 SPECIAL TOOL & TEST EQUIPMENT.....	20-20
20-00-00 CAUTIONS & WARNINGS.....	20-20

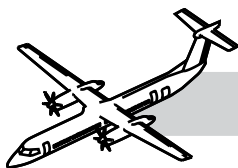


ILLUSTRATIONS

Figure	Title	Page
20-1	Pitot Static Probe Drain Hole Cleaning.....	20-3
20-2	Inspection of Control Cables.....	20-4
20-3	Permaswage Fittings	20-6
20-4	Sierracin/Harrison Quickfit and Harrison H-Fittings	20-6
20-5	Aeroquip Rynglok Unions	20-6
20-6	CryoFit and Cryolive Couplings	20-8
20-7	Deutsch Miniature End (DME) Fittings	20-8
20-8	Flexible Hoses	20-9
20-9	ESDS Labels.....	20-10
20-10	Conductive Dust Caps.....	20-10
20-11	Conductive Work Surface, Wrist Strap/Cord and Conductive Bag.....	20-11
20-12	Lubricants and Corrosion Inhibitors	20-12
20-13	Contact Enhancer.....	20-12
20-14	Geltek Tape.....	20-12
20-15	Degaussing Equipment	20-14
20-16	Lens Cap Assembly and Switch Light Bulb	20-14
20-17	Instrument Panel Integral Lamp Assembly	20-16
20-18	Vacuum Anchor System	20-16
20-19	Removal of Electrostatic Discharge Sensitive Devices	20-18

TABLES

Table	Title	Page
20-1	Approved Permanent and Non-Permanent Type Markers	20-2



CHAPTER 20

STANDARD PRACTICE



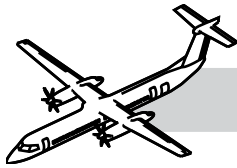
20 STANDARD PRACTICES
AIRFRAME

20-00-00 STANDARD PRACTICE

20-00-01 MARKING OF THE AIRCRAFT PARTS AND MATERIALS

INTRODUCTION

Markers are used to identify the parts, materials and assemblies of the aircraft. Identification markings that are done correctly will not cause damage to the function or the surface of the parts.



GENERAL

NOTES

If necessary, identification marks are put on titanium, plastics, composite material or painted surfaces, as follows:

- Identification marks are made on a tag or label
- The tag or label is attached to the surface of the part
- If the part is in a bag, the tag or label is attached to the bag.

The temporary identification marks are removed from the parts with a clean, lint-free cloth and water.

If it is necessary to put non-destructive testing (NDT) defect indication marks on painted surfaces, the approved temporary marker in Table 20-1 is used.

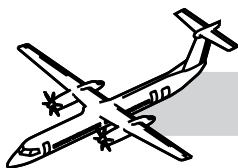
The marks are removed when no longer required by solvent cleaning.

Approved permanent markers are used for permanent identification on surfaces that are not titanium, plastics, composite materials or painted surfaces.

The permanent identification marks are removed from the parts by solvent cleaning.

Type	Point	Marker	Color - Part Number
Permanent (removed by solvent cleaning)	Thick	Marks-a-lot (Carter's)	Black - C85201
		Action Marker (Mark Tex Corp.)	Black - MK-DALO-B3
	Thin	Sharpie (Sanford's)	Black - 30001
Temporary (removed with water)	Thick	6000 (Faber Castell)	Black - 6001
	Thin	Vis-a-Vis (Sanford's)	Black - 1601
Temporary (removed by solvent cleaning) See Note 1.	-	"China" Marker	Dixon Ticonderoga Co., Phano No. 71, red.
NOTE 1: Acceptable for use on painted surfaces for marking defect indications revealed through non-destructive testing (NDT) only.			

Table 20-1. Approved Permanent and Non-Permanent Type Markers



MARKING OF THE AIRCRAFT PARTS AND MATERIALS WITH MARKERS

You can use markers to identify parts, materials and assemblies of the aircraft. Identification markings that are done correctly will not cause damage to the function or the surface of the parts.

CAUTION

DO NOT USE A PENCIL ON THE AIRCRAFT PARTS, MATERIALS, OR ASSEMBLIES.

CAUTION

DO NOT USE THE PERMANENT OR TEMPORARY MARKERS ON TITANIUM, PLASTICS, COMPOSITE MATERIALS, PAINTED SURFACES OR FABRIC AND UPHOLSTERY MATERIALS.

CAUTION

DO NOT WRITE DIRECTLY ON POLYETHYLENE, OR THIN PAPER BAGS WITH THE

PERMANENT OR TEMPORARY MARKERS, IF THE BAGS CONTAIN TITANIUM, PLASTICS, COMPOSITE MATERIALS, PAINTED PARTS OR FABRIC AND UPHOLSTERY MATERIALS, YOU CAN CAUSE DAMAGE TO THE PARTS.

If it is necessary to put identification marks on titanium, plastics, composite material or painted surfaces, make the identification marks on a tag or label and attach the tag or label to the part.

CLEANING THE DRAIN HOLE ON THE PITOT STATIC PROBES

Refer to Figure 20-1. Pitot Static Probe Drain Hole Cleaning.

Put a light on the drain hole of the probe and make sure that you can see the light in the pitot chamber through the pitot entry of the probe.

Use a smooth nylon wire (do not use a metallic rod or wire) with a diameter of 0.023 in (0.6mm) to clean the drain hole. Do not put more than 0.118 in (3 mm) of the nylon wire into the drain hole.

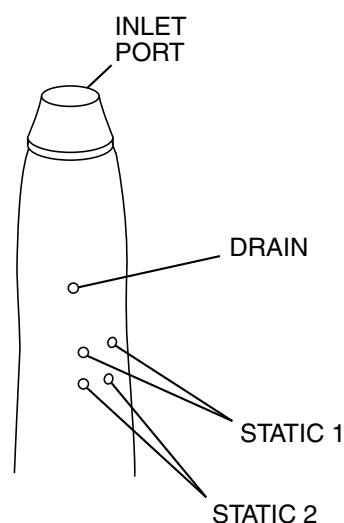
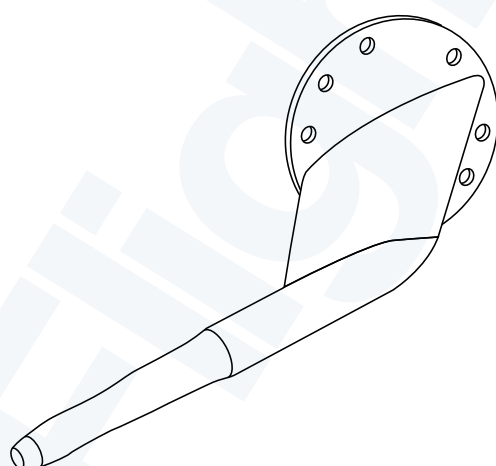


Figure 20-1. Pitot Static Probe Drain Hole Cleaning

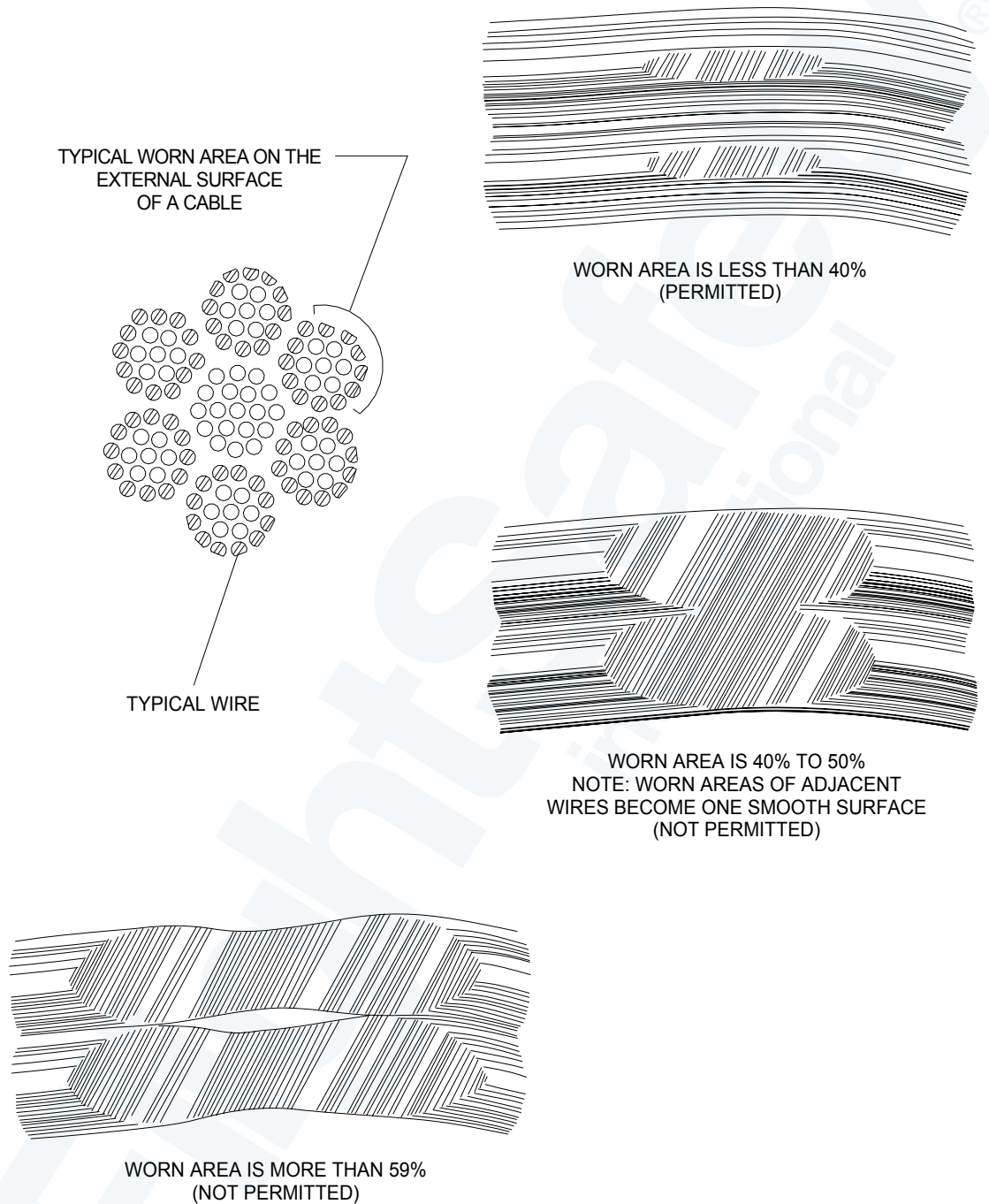
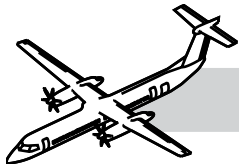
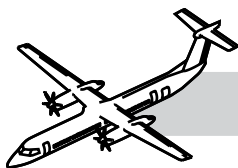


Figure 20-2. Inspection of Control Cables



INSPECTION OF SEVEN-BY-NINETEEN AND SEVEN-BY-SEVEN CONTROL CABLES

NOTES

Refer to Figure 20-2. Inspection of Control Cables.

The 7-by-19 cables have seven groups of wires with 19 wires in each group. The 7-by-7 cables have seven groups of wires with seven wires in each group. The flight control system, the landing gear system, and the doors use these types of cables. The control cables usually wear where the cables go through fairleads, or on pulleys and quadrants.

When carbon steel cables that are corrosion resistant are made, the cable strands are plated with tin and zinc and an internal lubricant is applied by the manufacturer. For carbon steel cables that are manufactured without an external jacket, the manufacturer also externally applies a lubricant to the cable. The lubricant will usually stay on the cable for its full service life.

DO NOT REMOVE THE LUBRICANT FROM THE EXTERNAL SURFACES OF THE CONTROL CABLES WITH VAPOR, STEAM, OR SOLVENTS.

Use a thick cloth to give protection to your hands when you examine the control cables for broken wires.

Examine the cables for the following and replace if the discrepancy exceed published AMM limits:

- Damage
- Wear
- Broken Strands
- Corrosion

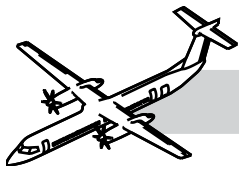


Figure 20-3. Permaswage Fittings



Figure 20-4. Sierracin/Harrison Quickfit and Harrison H-Fittings

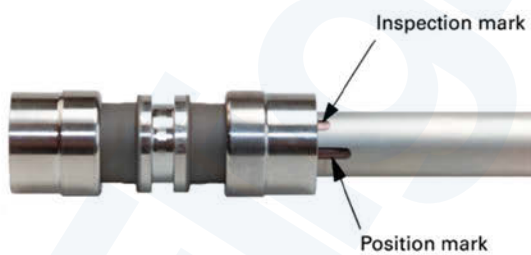
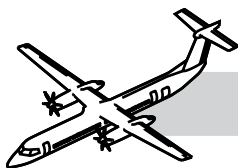


Figure 20-5. Aeroquip Rynglok Unions



GUIDELINES FOR RIGID LINE REPAIRS

Several AMM tasks address the repair of rigid lines based on their application and material construction. Repairs can be of a permanent or temporary nature.

Permanent Repairs

Permaswage Fittings

Refer to Figure 20-3. Permaswage Fittings.

This task procedure gives data for the repair of rigid fuel lines (only Aluminum 6061-T4, T6, Titanium and CRES tubes) with permaswage fittings. The procedure is for the repair of any fuel tube in and out of the fuel tanks.

Harrison H-Fittings

Refer to Figure 20-4. Sierracin/Harrison Quickfit and Harrison H-Fittings.

This maintenance practice gives the procedures and requirements necessary to use a Harrison H-Fitting to repair a damaged or defective hydraulic line. A Harrison H-Fitting is an assembly of a union, a coupling, a sleeve and a nut. The coupling assembly is suitable for use on hydraulic high pressure tubes of 3000 psi (20,700 kPa) pressure by using Titanium 3AL-2.5, CRES 21-6-9 and CRES 304 1/8 hard tubing. The repair can be used on any applicable hydraulic system tubing, except in hot sections, and can be accomplished with two end wrenches of suitable size.

Sierracin/Harrison Quickfit Fittings

Refer to Figure 20-4. Sierracin/Harrison Quickfit and Harrison H-Fittings.

This maintenance practice gives the procedures and requirements for the permanent repair of damaged or defective tubing when you use Sierracin/Harrison Quickfit fittings, straight (union), shaped (tee and 90 degree elbow), bulkhead (union, tee, 45 and 90 degree elbow) or adapter (Quickfit to flared and flareless) fittings.

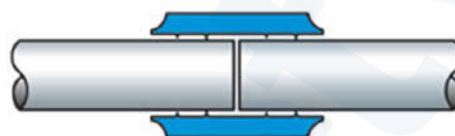
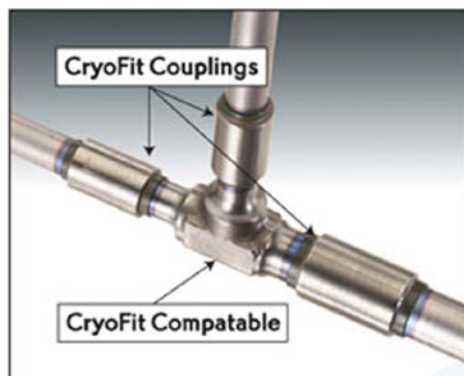
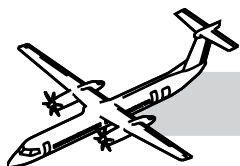
Aeroquip Rynglok Unions

Refer to Figure 20-5. Aeroquip Rynglok Unions.

This maintenance practice gives the data for the repair of hydraulic and pneumatic lines with Aeroquip Rynglok Fittings. Rynglok fittings are made from 6AL-4V titanium and can be used on aluminum, corrosion resistant steel (CRES) or titanium hydraulic tubes. These fittings are used on different tubes and pressure ratings to a maximum of 8000 psig (55200 kPa).

The fittings are swaged to the tube with a hydraulic swage tool that can be operated by a manual hydraulic power unit or automated air-driven hydraulic power unit. The aluminum tubes must have their specified heat treatment before you swage the fitting.

The tubes can be bent before or after the swage operation if tube bends are not less than 1 in (25.4mm) from tube ends.



Expanded couplings have an inside diameter slightly larger than the tube outside diameter

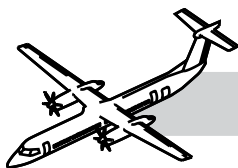


As the coupling warms and recovers it swagers on the tubing generating a highly reliable metal to metal seal

Figure 20-6. CryoFit and Cryolive Couplings



Figure 20-7. Deutsch Miniature End (DME) Fittings



CryoFit and Cryolive Couplings

Refer to Figure 20-6. CryoFit and Cryolive Couplings.

This maintenance practice gives the procedures for the permanent repair of damaged or defective tubing using CryoFit and Cryolive couplings made by Advance Metal Components Incorporated.

CryoFit-Cryolive fittings can be used on aluminum, CRES, stainless steel and titanium tubes.

The fittings are shipped in special insulated liquid nitrogen containers.

When correctly installed, the Cryo sleeve seals around the tube. Compression of the tubes and a permanent connection will occur.

When using these fittings, make sure the time the fittings are at room temperature is short as the fittings start to change their shape at room temperature.

Deutsch Miniature End (DME) Fittings

Refer to Figure 20-7. Deutsch Miniature End (DME) Fittings.

This maintenance practice gives the procedures and requirements for the permanent repair of damaged or defective tubing when you use

Deutsch Miniature End fittings series (DME). DME is a tube fitting that is mechanically attached to the end of a tube by axially swaging. The procedure involves swaging the fitting into the tube by using a Deutsch End fitting Tool (DET). This produces an interference fit and the tube results in a leak-free metal-to-metal seal.

The Procedure requires trained personnel to do the work for the best results and safe handling of the tooling. Training is provided by “Deutsch Metal Components”.

Repair of Hydraulic Lines Using Flexible Hoses

Refer to Figure 20-8. Flexible Hoses.

This standard practice gives the procedures and conditions for the temporary replacement and repair of damaged or defective rigid hydraulic tube using flexible hose assemblies. A total time of 5000 flight hours is allowed with flexible hose inspection each 1000 flight hours.

If necessary two or more flexible hoses connected to each other may be used however, they must be inspected every 50 hours and replaced with an approved permanent rigid tube or with one length before 1000 FH.

Flareless or flared type fittings, unions, adapters and elbows may be used. The task includes some of the applicable hardware for quick reference.



Figure 20-8. Flexible Hoses

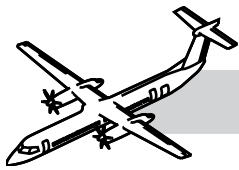
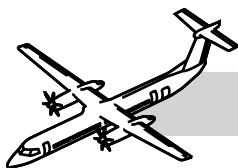


Figure 20-9. ESDS Labels



Figure 20-10. Conductive Dust Caps



TASK REMOVAL & INSTALLATION OF ELECTROSTATIC DISCHARGE SENSITIVE DEVICES

Provides instructions about how to identify and remove/install a device that is electrostatic discharge sensitive (ESDS). An ESDS device is a component that can be internally damaged by an electrostatic discharge. Two procedures are detailed in the tasks, one if the device has a metal case and the other if the device is a printed circuit board. Both of these devices will have a typical ESDC label (Refer to Figure 20-9).

Metal Case

If the ESDS device has a metal case once removed from its mounting tray do not touch any of the electrical pins or sockets and immediately install conductive dust caps (Black or Grey) (Refer to Figure 20-10).

Printed Circuit Board

If the device is a printed circuit board a conductive work surface, wrist strap/cord and conductive bag will be required.

Put the conductive work surface adjacent to the area where you will remove the printed-circuit board. Attach the grounding cord to the conductive work surface and a good grounding point on the aircraft structure. Attach the wrist strap to your wrist and the grounding cord of the wrist strap to a good grounding point on the aircraft structure. Remove the printed-circuit board and immediately put the printed-circuit board in the conductive bag. Seal the conductive bag and place an ESDS label on the bag (Refer to Figure 20-11).



Figure 20-11. Conductive Work Surface, Wrist Strap/Cord and Conductive Bag

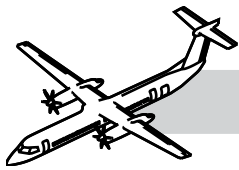


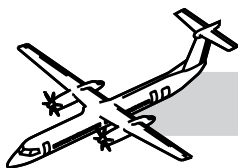
Figure 20-12. Lubricants and Corrosion Inhibitors



Figure 20-13. Contact Enhancer



Figure 20-14. Geltek Tape



Cleaning of Electrical Connectors, Switches or Relays

This maintenance practice is not applicable for vendor components, such as power lever module. For PW 150A engine wiring harness connector cleaning procedures refer to task 70-00-15-110-801.

Refer to Figure 20-12. Lubricants and Corrosion Inhibitors.

These procedures approve the use of lubricants for preventing corrosion and to remove moisture from electrical connectors, switches and relays.

- Use Super Corr-A, Super Corr-B, ACF-50 or LPS 2 as a lubricant and corrosion inhibitor for electrical connectors or relay base contacts.
- Use ACF-50 and LPS 2 to remove moisture as well as to prevent corrosion on relays and switches.

Use Stabilant 22 or 22A to improve the conductivity of contacts in connector pins, card edge connectors, signal switches and relay base pins. Stabilant 22 is undiluted and not recommended to apply directly on connector pins or sockets because of poor flow of material. Dilute Stabilant 22 with alcohol. Stabilant 22A is diluted with alcohol (4:1) and recommended for connectors and relay pins (Refer to Figure 20-13).

Use Geltek sealant tape for electrical connectors to improve their corrosion resistance (Refer to Figure 20-14).

Cleaning of the Wiring Harness Connectors (Engine)

The maintenance procedure is for the cleaning of the engine wiring harness electrical connectors. Any grease must be removed from the connectors using Acetone, Petroleum solvent or a Cleaning solvent. See PSM 1-84-18, Consumable Products Manual for details of the approved products.

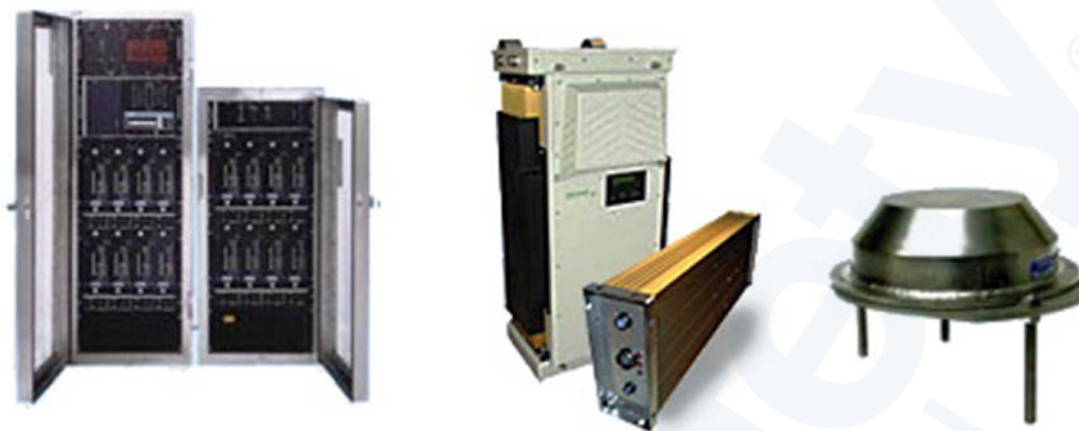
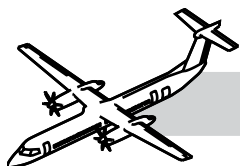


Figure 20-15. Degaussing Equipment

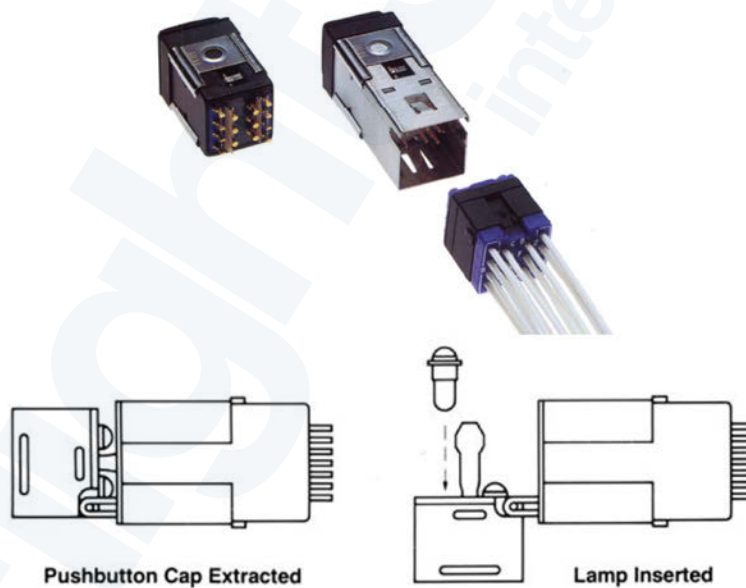
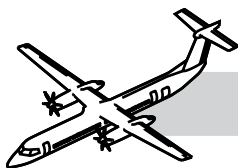


Figure 20-16. Lens Cap Assembly and Switch Light Bulb



Demagnetization of the Aircraft Components and/or Structures

Refer to Figure 20-15. Degaussing Equipment.

Steel/iron components may become magnetized when placed in the presence of a large magnetic field or when a large electrical currents (lighting) are conducted through them. When components are magnetized, they should be degaussed to prevent system errors from occurring. The strength of magnetic field is measured in Gauss units.

Degaussing is the process of removing residual magnetism from a component. The process consists of placing the magnetized component within a larger magnetic field of opposite polarity and slowly reducing the magnitude of the field strength to zero.

Replacement of the Lens Cap Assembly and Switch Light Bulb

Refer to Figure 20-16. Lens Cap Assembly and Switch Light Bulb.

This task describes the procedure for the replacement of lens cap assembly and switch light bulbs used in the illuminated switches.

Replace the bulb:

- Hold the grip and remove the lens cap assembly from the switch assembly.
- Remove the lamp carrier from the lens cap assembly.
- Remove the bulb from the lamp carrier.
- Insert a new bulb in the lamp carrier and install it in the lens cap assembly.
- Install the lens cap assembly on the switch assembly.

Replace the lens cap assembly:

- Hold the grip and remove the lens cap assembly from the switch assembly.
- Disconnect the wire bail and remove the lens cap assembly.
- Connect the wire bail to the new lens cap assembly.
- Install the lens cap assembly on the switch assembly.

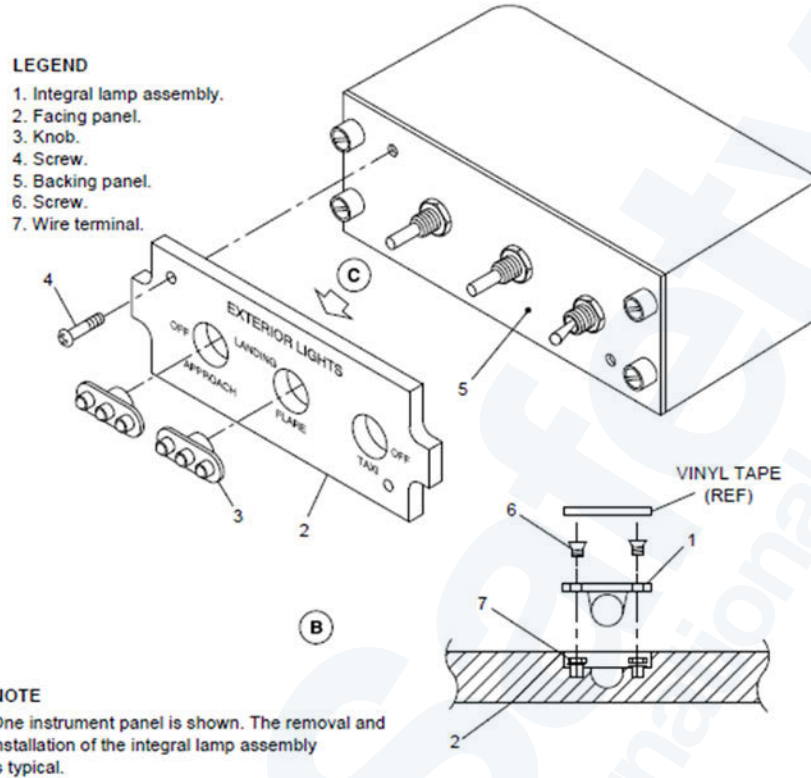
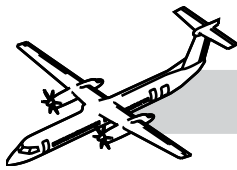
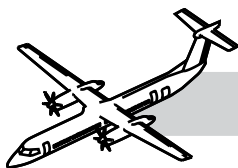


Figure 20-17. Instrument Panel Integral Lamp Assembly



Figure 20-18. Vacuum Anchor System



Electrical Test of the Aircraft Wiring

The task procedures are for the electrical tests of the aircraft wiring. These procedures include connector inspections, continuity and isolation tests (to verify the integrity of the electrical wiring), with an ohmmeter and a mega ohmmeter (high voltage insulation tester or Megger), DC and AC voltage checks and bonding tests.

The task includes the following procedures:

- Inspection of connectors for:
- Corrosion
- Mechanical damage
- Evidence of water ingress or any other liquid in the pin/socket area
- Burned pin/socket or arcing damages
- Bent pins
- Loose connector parts

Installation of the Aircraft Antennas & Application of Sealant to Antenna Connectors

These tasks provide instructions for the installation, (including torquing and sealing of the aircraft antennas and their connectors.

Removal/Installation of the Instrument Panel Integral Lamp Assembly

Refer to Figure 20-18. Vacuum Anchor System.

The task procedure is for the removal & installation of the instrument panel integral lamp assemblies from the facing panels. The procedure is applicable to all the integral lamp assemblies installed with different facing panels in the flight compartment.

VACUUM ANCHOR SYSTEM

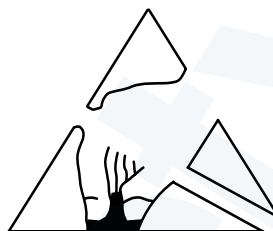
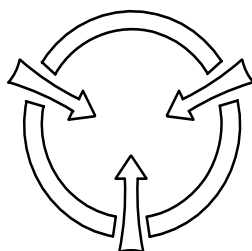
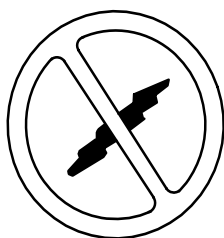
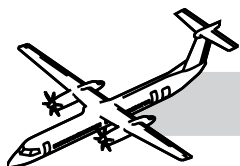
Refer to Figure 20-18. Vacuum Anchor System.

The vacuum anchor system provides a person-rated fall arrest anchorage point for maintenance crews as they perform maintenance and servicing of the aircraft. The system consists of a vacuum anchor assembly with an on-board gas (Nitrogen) bottle, an energy absorbing lanyard and a full body harness. This equipment is listed in the Illustrated Tools and Equipment Manual (ITEM) as P/N: GSB2000018.

NOTE

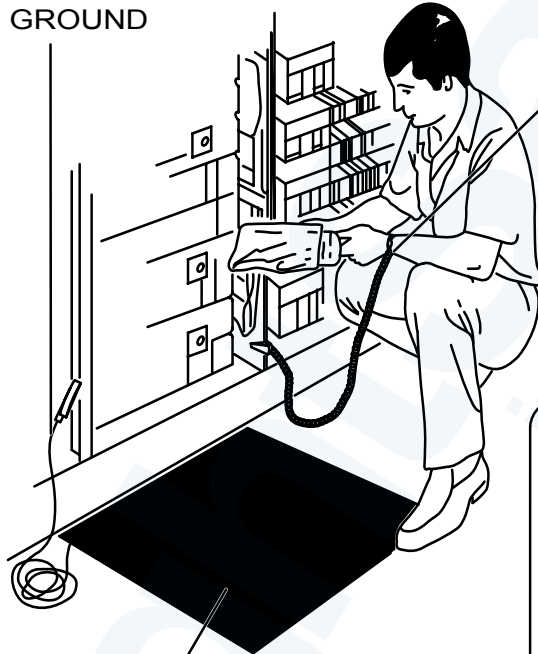
The vacuum anchor system is a part of a complete fall protection system. The user must read and follow the manufacturer's Instructions for each component of the system.

The vacuum anchor system is designed for use by persons with a combined weight (clothing, tools, etc.) of no more than 310 lbs (140.6 kg). Only one person fall arrest system may be connected to a single vacuum anchor.

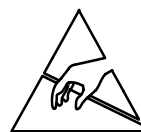


CONNECT THE
GROUND CORD TO
THE WORK SURFACE
AND THE ALLIGATOR
CLIP TO A GOOD
GROUND

WRIST STRAP-
CONNECT THE
ALLIGATOR CLIP
TO ANY GOOD
GROUND



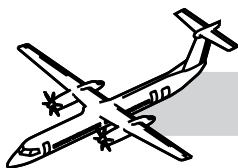
10 MIL CONDUCTIVE
WORK SURFACE



ATTENTION

**Static Sensitive
Devices**
Handle Only at
Static Safe Work
Stations
Reusable Container
Do not Destroy

Figure 20-19. Removal of Electrostatic Discharge Sensitive Devices



20-30-00 REMOVAL OF ELECTROSTATIC DISCHARGE SENSITIVE DEVICES

INTRODUCTION

Instructions are given about how to identify and remove a device that is electrostatic discharge sensitive (ESDS). An ESDS device is a component that can be damaged by an electrostatic discharge.

GENERAL

Refer to Figure 20-19. Removal of Electrostatic Discharge Sensitive Devices.

An ESDS device is removed as follows:

- The electrical system power is removed from the ESDS device
- The necessary panels are removed to get access to the equipment
- Ensure that the device has a typical ESDS label.

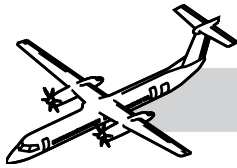
If the ESDS device has a metal case, the following is done:

- The electrical wire connectors are disconnected from the ESDS device
- Conductive dust caps are installed on the electrical connectors
- The clamps that attach the ESDS device to the mounting tray are loosened
- The ESDS device is removed from the mounting tray.

If the ESDS device is a printed-circuit board, the following is done:

- The conductive work surface is put adjacent to the area where the printed-circuit board is removed

- The grounding cord is attached to the conductive work surface and a good grounding point on the aircraft structure
- The wrist strap is attached to your wrist and the grounding cord of the wrist strap is attached to a good grounding point on the aircraft structure
- The printed-circuit board is removed and immediately put in the conductive bag
- The conductive bag is sealed with a cotton cord or an ESDS label
- Ensure that there is an ESDS label on the bag
- The printed-circuit board and the bag is put in a rigid container before it is sent to the shop to prevent damage to the conductive bag
- The wrist strap and conductive work surface are disconnected from the grounding points.



20-00-00 SPECIAL TOOL & TEST EQUIPMENT

- Commercially available conductive work surface
- Commercially available static wrist strap with ground cord; 1 MΩ resistor
- Commercially available static ground cord with clip; 1 MΩ resistor.

20-00-00 CAUTIONS & WARNINGS

NOTE

If the ESDS device is a printed-circuit board it will have a STATIC SENSITIVE label.

NOTE

Conductive dust caps should be black or gray.

NOTE

Make sure that you do not touch the pins in the electrical connectors on the ESDS device.

CAUTION

To prevent possible damage:

- Do not use a pencil on the aircraft parts, materials, or assemblies.
- Do not use the permanent or temporary markers on titanium, plastics, composite materials, or painted surfaces.
- Do not write directly on polyethylene, or thin paper bags with the permanent or temporary markers. If the bags contain titanium, plastics, composite materials, or painted parts, damage can occur to the parts.